DEPI - Digital Egypt Pioneers Initiative

UK Train Rides

Data Analysis & Forecasting Project

Phase 1: Project Planning, Literature Review and Requirements

TRACK **Professional Data Analyst**

Yousef Mohammed Elgabry	**Adham Reda Ghazy**	**Mohamed Gamal Emam**
Youssef Sherif Mohamed	**Ahmed Nasser Abdelhafez**	**Mohamed Ibrahim Elgafrawy**

Project Idea

UK Train Rides is a data analysis and forecasting project on UK train ticket sales and journeys.

What we analyze

Ride demand, revenue, punctuality, delays, refunds and ticket-class usage.

What we forecast

Next month's rides, daily revenue for each day, and demand per ticket class using machine learning.

What we deliver

A clean dataset, analysis answers, forecasts, a dashboard, a final report and presentation.

The data

Railway ticket dataset (railway.csv) with its data dictionary: ticket purchases and journeys from January to April 2024, including price, ticket class and type, route, delays, cancellations and refund requests.

Objectives

1. Build a data model and a clean, analysis-ready dataset with a documented notebook
2. Define all analysis questions the data can answer for decision makers
3. Build machine learning models to forecast rides, daily revenue and ticket-class demand (May 2024)
4. Build a dashboard that visualizes the answers to all analysis questions
5. Deliver a final report and presentation

Project Plan

Week 1
17 Oct - 6 Nov

Data model, cleaning, preprocessing

Week 2
7 - 13 Nov

Analysis questions

Week 3
14 - 22 Nov

Forecasting (machine learning)

Week 4
23 Nov - 4 Dec

Dashboard, final report and presentation

DEPI deadlines

16 Oct
Planning, Literature Review, Requirements

6 Nov
System Analysis & Design

30 Nov
Implementation (source code)

4 Dec
Final presentation, testing, reports

Team & Roles

Yousef Mohammed Elgabry

Team Leader & Dashboard

Mohamed Ibrahim Elgafrawy

Dashboard

Youssef Sherif Mohamed

SQL & Data Model

Mohamed Gamal Emam

SQL & Data Model

Adham Reda Ghazy

Python

Ahmed Nasser Abdelhafez

Python

Whole team: forecasting (machine learning), testing and documentation

Risks & KPIs

Main risks and how we handle them

Short data history (about 4 months)

Use the weekly pattern and recent trend; compare with a simple baseline

Unusual first-of-month volumes

Flag those days and test with and without them

Inconsistent labels in the data

Documented mapping during cleaning

Mock data

State clearly that results describe this dataset

Success measures (KPIs)

- Deliverables on time
- Data issues resolved and documented
- 100% of analysis questions answered in the dashboard
- Forecast error lower than a simple baseline
- All 3 forecasts delivered
- Reproducible notebook and scripts

Literature Review

Dashboard projects (5)

Public projects on the same dataset. Strong descriptive analysis, but no forecasting.

Academic work (3)

Rail demand forecasting studies. Use ARIMA, neural networks and more, but need years of data.

Our project

Combines SQL data model, clean data, machine learning forecasts and a dashboard on a short history.

What we take from them

- Common business questions: routes, peak times, revenue by ticket type and class, punctuality, refunds
- Explicit cleaning checks (formats, duplicates, blanks)
- Several error metrics, a baseline, and the weekly pattern in daily ticket demand

Requirements

Stakeholders

- Decision makers (operations, commercial, planning)
- Lecturer / DEPI evaluators
- Project team
- Data owner / provider
- Dashboard end users

Key functions

- Load data into a relational model
- Clean and preprocess the data
- Answer analysis questions with visuals
- Forecast rides and daily revenue
- Forecast demand per ticket class
- Dashboard with filters
- Final report and presentation

Quality

- Performance: fast dashboard filters
- Security: no personal data, no credentials in the repository
- Usability: clear and easy to navigate
- Reliability: forecasts validated on held-out data

Next Steps

By 6 Nov

System Analysis & Design

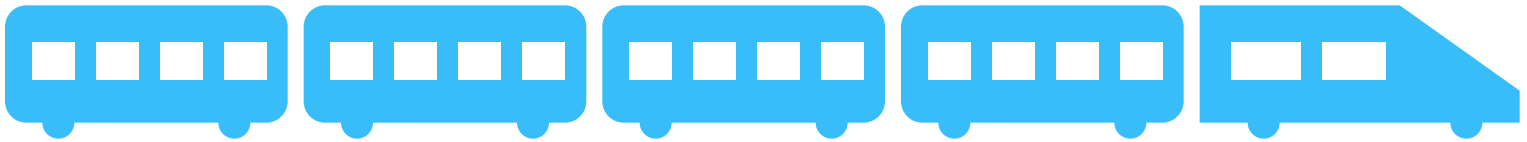
ER diagram, database schema, UI/UX wireframes, diagrams

By 30 Nov

Implementation

Source code on GitHub, cleaned data, forecasting models, dashboard

By 4 Dec

Final delivery

Testing, final report, presentation and demo